

Colligative properties of electrolyte

21. (b) NaCl gives maximum ion hence it shows lowest freezing point

22 Explanation:

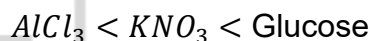
Depression in freezing point:

$$\Delta T_f = iK_f m$$

Greater the van't Hoff factor i , greater the depression in freezing point, hence lower the freezing point.

- Glucose $\rightarrow$ non-electrolyte $\rightarrow i = 1$
- $\text{KNO}_3 \rightarrow \text{K}^+ + \text{NO}_3^- \rightarrow i = 2$
- $\text{AlCl}_3 \rightarrow \text{Al}^{3+} + 3\text{Cl}^- \rightarrow i = 4$

Thus freezing point decreases in order:



(a) $\text{AlCl}_3 < \text{KNO}_3 < \text{Glucose}$

23. (b) Lesser the number of particles in solution. Lesser the depression in freezing point, i.e. higher the freezing point.

24. (c) BaCl_2 gives maximum ion hence it shows maximum depression in freezing point.

25. Explanation:

Depression in freezing point:

$$\Delta T_f = iK_f m$$

All are non-electrolytes, so $i = 1$.

All solutions have same molality $0.1m$, therefore all produce the same depression in freezing point.

Hence all have the same freezing point.

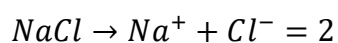
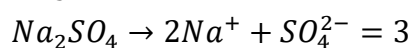
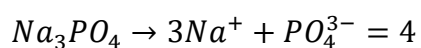
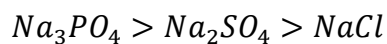
Answer: All have the same freezing point

26. (d) We know that lowering of freezing point is a colligative property which is directly proportional to the number of particles formed by one mole of compound therefore $0.1M \text{Al}_2(\text{SO}_4)_3$ solution will have minimum freezing point.



27. (a) $Al_2(SO_4)_3$ gives maximum ion hence its gives lowest freezing point.

28. (b) Colligative property in decreasing order



29. (d) $K_4[Fe(CN)_6]$ gives maximum ion. Hence it have lowest vapour pressure.

